

Views for Security

Exercise 1

```
CONNECT C##STUDENT_A_1701497/pass1234;

/*My oracle notes and trials
To show the tables created by the user logged in use the statement*/
SELECT TABLE_NAME FROM USER_TABLES;

-- This will only show the tables the user created or rather owned by the user

-- Viewing the tables Accessible by the current user ... use the statement

SELECT TABLE_NAME FROM ALL_TABLES

--LAB 4

--1
/*SELECT TABLE_NAME FROM ALL_TABLES ORDER BY TABLE_NAME; */

CONNECT C##STUDENTB_1701497/pass1234;
--Viewing all the tables accessible to the user which shows student B has view
privileges to
-- student A
SELECT TABLE_NAME , OWNER FROM ALL_TABLES WHERE OWNER='C##STUDENT_A_1701497' ORDER BY
TABLE_NAME;

-- Can also view student A schema
Select * from C##STUDENT_A_1701497.Sales;

-- Revoke Viewing privileges
-- I could use the dba admin account or use student A which also have authority to
revoke access to
-- another user since it is the owner

CONNECT C##DBADMIN/Password1;

REVOKE SELECT ON C##STUDENT_A_1701497.SALES FROM C##STUDENTB_1701497;
-- Don't need to specify any more privileges since only select priv was granted to
Student B

CONNECT C##STUDENTB_1701497/pass1234;

Select * from C##STUDENT_A_1701497.Sales;

--2

CONNECT C##STUDENT_A_1701497/pass1234;

CREATE VIEW VIEW_REORDER_1701497 AS
SELECT * FROM SALES WHERE QUANTITY < 5 ;

/*ERROR at line 1:
ORA-01031: insufficient privileges -- user doesn't have privilege to create
view so have to grant it by an admin */
```

```
CONNECT C##DBADMIN/Password1;

GRANT CREATE VIEW TO C##STUDENT_A_1701497;

/*Create view privileges was grant to student A */

CONNECT C##STUDENT_A_1701497/pass1234;

CREATE VIEW VIEW_REORDER_1701497 AS
  SELECT * FROM SALES WHERE QUANTITY < 5 ;

---3

CONNECT C##STUDENTB_1701497/pass1234;

SELECT TABLE_NAME , OWNER FROM ALL_TABLES WHERE OWNER='C##STUDENT_A_1701497'
ORDER BY TABLE_NAME;

/*This table shows all the tables accessible by the current user which can
also specify the owner in this case student A of sales where no results was
found meaning it doesn't (STUDENT B ) has access to its tables or schema*/

--4

CONNECT C##STUDENT_A_1701497/pass1234;

GRANT SELECT ON VIEW_REORDER_1701497 TO C##STUDENTB_1701497;

--Obervation

CONNECT C##DBADMIN/Password1;

SELECT * FROM DBA_TAB_PRIVS WHERE GRANTEE ='C##STUDENTB_1701497';
/*Shows all the grants on the objects in the database where specified
the grantee which is Student B that shows the granted privileges to the user*/

/*The dba table privileges confirm the privileges assign to the grantee as
specified above and snapshot*/

--5
CONNECT C##STUDENTB_1701497/pass1234;
SELECT * FROM C##STUDENT_A_1701497.VIEW_REORDER_1701497;

/*Statement ran successfully .. enter though there isn't any record under 5
quantity entered by me it returns "no records found" which is correct and for
more validation the user have access the database did not return table or object
doesn't
  exist concluding the user does have access to the view created*/

--6
```

```
SELECT * from C##STUDENTA_1701497.Sales;

/*As expected the table or view does not exist since only the created view
was granted to the user to view therefore unable to view the ful sales table*/

--ORA-00942: table or view does not exist

/*

--This can be resolve by granting viewing privilege on the sales table either
using the admin user or the owner of the table account
CONNECT C##DBADMIN/Password1;
GRANT SELECT ON C##STUDENTA_1701497.Sales TO C##STUDENTB_1701497;

*/
```

Exercise 2

```
-- EXERCISE 2
```

```
--Connect back to the admin account
CONNECT C##DBADMIN/Password1;

--Enable auditing
/*PFILE stands for the parameter file. It is a text file which can be modified
by a text editor. With the help, PFILE Oracle server start an Oracle Instance.*/

ALTER SYSTEM SET audit_trail=db SCOPE=SPFILE;

/*For the auditing features to take effect the server will need to be restarted*/

CONNECT SYS/Backtrack#r3 AS SYSDBA;
/*When trying to shutdown the database instance it return
error insufficient privileges.. Possible the created user cannot or has the
privileges.

Logged into the 'sys as sysdba' to resolve this error*/

SHUTDOWN IMMEDIATE

STARTUP

/*Auditting features are now in effect*/

SHOW PARAMETER AUDIT_TRAIL

--1

AUDIT SELECT TABLE, UPDATE TABLE, INSERT TABLE, DELETE TABLE BY C##STUDENTA_1701497
BY ACCESS;
```

```

/*Used BY ACCESS mean if you want Oracle Database to write one record for each
audited
statement and operation.*/

/* "DBA_STMT_AUDIT_OPTS" Table use to see and Describes current system auditing
options across the system and by user */
SELECT * FROM DBA_STMT_AUDIT_OPTS;

DESCRIBE DBA_STMT_AUDIT_OPTS;

--USER_NAME - User name if by user auditing
--PROXY_NAME - Name of the proxy user which is performing an operation for the
client
--AUDIT_OPTION - Name of the system auditing option
---SUCCESS - Mode for WHENEVER SUCCESSFUL system auditing
--FAILURE - Mode for WHENEVER NOT SUCCESSFUL system auditing

--2

/*IF wanted to see the table creation would have used "AUDIT TABLE BY
C##STUDENT_1701497"
which would have been a DDL but not using this lab exercise*/

CONNECT C##DBADMIN/Password1;

/*Note: Unable to create table as expected since the privilege in role '
c##labclassA'
was revoked in the previous lab...
i will just grant back the privilege to the user */

--grant privilege back
GRANT CREATE ANY TABLE TO C##STUDENT_1701497;

CONNECT C##STUDENT_1701497/pass1234;

CREATE TABLE FOOD (
ID NUMBER,
Name VARCHAR(15)
);

--INSERT
INSERT ALL
INTO FOOD VALUES (1001, 'Apple')
INTO FOOD VALUES (1002, 'Mango')
SELECT * FROM DUAL;

--SELECT
SELECT * FROM FOOD;

--UPDATE
UPDATE FOOD
SET NAME = 'Chicken'
WHERE ID = 1002;

--DELETE
DELETE FROM FOOD
WHERE ID = 1001;

```

```

DESCRIBE DBA_AUDIT_TRAIL

/*Username: Oracle Username.
• Terminal: Machine that the user performed the action from.
• Timestamp: When the action occurred.
• Object Owner: The owner of the object that was interacted with.
• Object Name: The name of the object that was interacted with.
• Action Name: The action that occurred against the object. (INSERT,
UPDATE, DELETE, SELECT, EXECUTE)
*/

SELECT USERNAME, TERMINAL, TIMESTAMP, OWNER, OBJ_NAME, ACTION_NAME from
DBA_AUDIT_TRAIL
WHERE USERNAME='C##STUDENTA_1701497' AND OBJ_NAME = 'FOOD' ORDER BY TIMESTAMP;

/*DML statements were recorded in the DBA_AUDIT_TRAIL system table*/

--3

CONNECT SYS/Backtrack#r3 AS SYSDBA

NOAUDIT SELECT TABLE, UPDATE TABLE, INSERT TABLE, DELETE TABLE BY
C##STUDENTA_1701497;

CONNECT C##STUDENTA_1701497/pass1234;
--INSERT
INSERT ALL
INTO FOOD VALUES (1003, 'Cake')
INTO FOOD VALUES (1004, 'Bulla')
SELECT * FROM DUAL;

--SELECT
SELECT * FROM FOOD;

--UPDATE
UPDATE FOOD
SET NAME = 'Cheery Cake'
WHERE ID = 1003;

--DELETE
DELETE FROM FOOD
WHERE ID = 1004;

SELECT USERNAME, TERMINAL, TIMESTAMP, OWNER, OBJ_NAME, ACTION_NAME from
DBA_AUDIT_TRAIL
WHERE USERNAME='C##STUDENTA_1701497' AND OBJ_NAME = 'FOOD' ORDER BY TIMESTAMP;

/*There wasn't any update audit entries which means it was disable successfully*/

```

Exercise 3

```
-- EXERCISE 3

--1

--inserted additional records help test the queries for results
-----
CONNECT C##STUDENT_1701497/pass1234;

INSERT ALL
INTO SALES VALUES ('D004', 'Cake', 50, 150)
INTO SALES VALUES ('E004', 'Melon', 1000, 3)
SELECT * FROM DUAL;

-----

-- Connect back to SYS
CONNECT C##DBADMIN/Password1;

BEGIN
DBMS_FGA.add_policy(
object_schema => 'C##STUDENT_1701497',
object_name => 'SALES',
policy_name => 'QUANTITY_CHK_AUDIT',
audit_condition => 'QUANTITY > 100',
audit_column=> 'QUANTITY');
END;
/
/*Results: PL/SQL procedure successfully completed.*/

/* DBA_FGA_AUDIT_TRAIL system table displays All audit records for fine-grained
auditing.*/

--2

SELECT * FROM DBA_FGA_AUDIT_TRAIL

/*Table has no data as yet*/

DESCRIBE DBA_FGA_AUDIT_TRAIL

-- List of some fields in the DBA_FGA_AUDIT_TRAIL and what they store
/*
DB_USER - Database username who executed the query
SCN - System change number (SCN) of the query
SQL_TEXT -SQL text of the query
TIMESTAMP - Date and time of the query in the local database session time zone
POLICY_NAME - Name of the Fine-Grained Auditing Policy
*/

--3

CONNECT C##STUDENT_1701497/pass1234;
```

```

SELECT * FROM SALES WHERE QUANTITY < 8 ;

--4

CONNECT C##DBADMIN/Password1;

SELECT DB_USER, SCN, SQL_TEXT, TIMESTAMP, POLICY_NAME FROM DBA_FGA_AUDIT_TRAIL;

/*As expected the specified criteria of the fined grained auditing policy
only audit the statements over 100 in quantity in the sales table*/

--5

CONNECT C##STUDENT_1701497/pass1234;

SELECT * FROM SALES WHERE QUANTITY > 10 ;

--6

CONNECT C##DBADMIN/Password1;
SELECT DB_USER, SCN, SQL_TEXT, TIMESTAMP, POLICY_NAME FROM DBA_FGA_AUDIT_TRAIL;

/*The system table now shows a record of the audit of the Sql query that included
quantity over
10 which would also fall in the group of quantity over 100 therefore activating or
firing the audit */

```

Exercise 4

```

-- EXERCISE 4

--1

CONNECT C##STUDENT_1701497/pass1234;

DESC SALES

/*a front slash is needed at end of the procedure statment  ' / ' */

CREATE OR REPLACE PROCEDURE InsertData_Sales(
    ID IN VARCHAR2 (25)
, PRODUCT IN VARCHAR2 (25)
, UNITCOST IN FLOAT
, QUANTITY IN NUMBER
) AS
BEGIN
    INSERT INTO SALES VALUES (ID, PRODUCT, UNITCOST, QUANTITY);
END InsertData_Sales;
/

/*The procedure is unable to create due to insufficient privilege*/

--Connect back to admin to grant create procedure to user
CONNECT C##DBADMIN/Password1;

```

```
GRANT CREATE PROCEDURE TO C##STUDENTA_1701497;
```

```
-----
```

```
CONNECT C##STUDENTA_1701497/pass1234;
```

```
--Attempted created the procedure but was encountered variety compilation errors in  
snapshots
```

```
SHOW ERROR
```

```
CREATE OR REPLACE PROCEDURE InsertData_Sales (  
    S_ID SALES.SALES_ID%TYPE,  
    Prod SALES.PRODUCT%TYPE,  
    UCost SALES.UNITCOST%TYPE,  
    Quant SALES.QUANTITY%TYPE )  
IS  
BEGIN  
    INSERT INTO SALES (SALES_ID,PRODUCT,UNITCOST,QUANTITY)  
    VALUES (S_ID, Prod, UCost, Quant);  
    COMMIT;  
END;  
/
```

```
/*Found the problem there should not be a common for the last declaration in parameter  
which was at the end of quantity which the help of the SHOW ERROR command was able to  
resolved the implications also replace "AS" to "IS" and removed the proc name from  
Infront the
```

```
END statement and also used an approach of matching the datatype of the parameter  
variables  
to those of the table type for precision.
```

```
Both have the same meaning in code standard 'IS' used when proc is nested and  
if standalone then 'AS' will be used other than that they both have same meaning  
*/
```

```
SELECT OBJECT_NAME FROM USER_PROCEduRES;
```

```
--2
```

```
CONNECT C##STUDENTB_1701497/pass1234;
```

```
INSERT INTO C##STUDENTA_1701497.SALES  
VALUES ('F006','Breadfruit',10,9);
```

```
/*Table or view does not exist
```

```
Note: Insert privileges and select was revoke from this user accessing Student A  
schema table*/
```

```
--3
```

```
-- In the positional notation that specify the actual parameter the same order as the  
formal parameter
```

```
-- Make sure the statement is wrap in begin and end
```

```
EXECUTE C##STUDENTA_1701497.InsertData_Sales('G008','StoreChicken',12,24);
```



```

/*The user does not have privilege to execute store procedure*/

CONNECT C##DBADMIN/Password1;

--Use the DBA table privileges system table
SELECT PRIVILEGE FROM DBA_TAB_PRIVS WHERE GRANTEE = 'C##STUDENTB_1701497';

--4

GRANT EXECUTE ON C##STUDENTA_1701497.InsertData_Sales TO C##STUDENTB_1701497;

/*Grant was successful*/

--Verifying use the DBA table privileges

SELECT PRIVILEGE FROM DBA_TAB_PRIVS WHERE GRANTEE = 'C##STUDENTB_1701497';

/*EXECUTE privilege on Student B has now been added*/

--5

CONNECT C##STUDENTB_1701497/pass1234;

EXECUTE C##STUDENTA_1701497.InsertData_Sales('G008', 'StoreChicken', 12, 24);

/*PL/SQL procedure successfully completed.*/

CONNECT C##STUDENTA_1701497/pass1234;

SELECT * FROM SALES;

--6
--redundant but connect to student A
CONNECT C##STUDENTA_1701497/pass1234;

CREATE OR REPLACE PROCEDURE sp_time_window_1701497 (
  S_ID SALES.SALES_ID%TYPE,
  Prod SALES.PRODUCT%TYPE,
  UCost SALES.UNITCOST%TYPE,
  Quant SALES.QUANTITY%TYPE )
IS
BEGIN
  IF ((to_char(sysdate, 'mm')) = '03') THEN
    INSERT INTO SALES (SALES_ID, PRODUCT, UNITCOST, QUANTITY)
    VALUES (S_ID, Prod, UCost, Quant);
  ELSE
    dbms_output.put_line('Can Only insert during the month of March.. oh hail
stackoverflow');
  END IF;
COMMIT;

```

```

END;
/
/*Noted: The DBMS_OUTPUT package enables you to send messages from stored
procedures, packages, and triggers. The package is especially useful for
displaying PL/SQL debugging information.*/

--7

CONNECT C##STUDENTB_1701497/pass1234;

EXECUTE C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);

/*ORA-06550: line 1, column 7:
PLS-00201: identifier 'C##STUDENTA_1701497.SP_TIME_WINDOW_1701497' must be
declared
ORA-06550: line 1, column 7:
PL/SQL: Statement ignored*/

/*Student A does not have execute privilege which proven from previous executing
of procedure to resolve grant execute privilege*/

CONNECT C##DBADMIN/Password1;

GRANT EXECUTE ON C##STUDENTA_1701497.sp_time_window_1701497 TO C##STUDENTB_1701497;

/*Grant succeeded*/

CONNECT C##STUDENTB_1701497/pass1234;

EXECUTE C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);

/*PL/SQL procedure successfully completed*/

```

Exercise 5

```

--EXERCISE 5

--1

CONNECT C##STUDENTA_1701497/pass1234;

CREATE TABLE Purchases(
  p_id varchar2(25),
  product varchar2(25),
  quantity Number,
  unitcost Float,
  total_cost Float
);

-- Uses dault date when date has been modified on table to system date
CREATE TABLE Purchases_log(
  eventdate date default SYSDATE,
  p_id varchar2(25),

```

```

description varchar2(50)
);

--2
/*There was an error relating to privilege in first attempt creating the trigger
will have to grant create trigger to this user by admin*/

CONNECT C##DBADMIN/Password1;

GRANT CREATE TRIGGER TO C##STUDENT_1701497;

--Grant succeeded.
-----

CONNECT C##STUDENT_1701497/pass1234;

CREATE OR REPLACE TRIGGER Purchases_AUDIT
BEFORE INSERT ON Purchases
REFERENCING NEW AS NEW OLD AS OLD
FOR EACH ROW
BEGIN

INSERT INTO Purchases_log(p_id, description)
VALUES(:NEW.p_id, 'INSERTING INTO PURCHASES');

END;
/

--Trigger created
----^ remember
/*Triggers are stored under the ALL_TRIGGERS (USING ADMIN) or USER_TRIGGERS owned by
the user*/

Select TRIGGER_NAME from USER_TRIGGERS;

--3

INSERT INTO PURCHASES(p_id,product,quantity,unitcost)
VALUES ('P001', 'Mango', 2, 20);

SELECT * FROM PURCHASES;
SELECT * FROM Purchases_log;

-- They both are populated only the total cost is empty

--4
--Use after update since the data would be in table after to perform calculation

CREATE OR REPLACE TRIGGER Update_Purchase_Audit
AFTER UPDATE ON Purchases
REFERENCING NEW AS NEW OLD AS OLD
FOR EACH ROW
BEGIN

INSERT INTO Purchases_log(p_id, description)
VALUES(:NEW.p_id, 'UPDATED TOTAL COST INTO PURCHASES');

END;
/

```

```
--Created
```

```
--5
```

```
UPDATE Purchases
  SET total_cost = ((quantity * unitcost) + ((quantity * unitcost) * 0.165) )
  WHERE p_id = 'P002';

SELECT * FROM PURCHASES;
SELECT * FROM Purchases_log;
```

```
--6
```

```
CREATE OR REPLACE TRIGGER Purchases_AUDIT
  BEFORE DELETE ON Purchases
  REFERENCING NEW AS NEW OLD AS OLD
  FOR EACH ROW
  BEGIN

  INSERT INTO Purchases_log(p_id, description)
  VALUES(:NEW.p_id, 'RECORD DELETED FROM PURCHASES');

  END;
/
```

```
DELETE FROM PURCHASES WHERE p_id = 'P001';

SELECT * FROM PURCHASES;
SELECT * FROM Purchases_log;
```

```
--SUCCESSFUL
```

Exercise 6

```
--EXERCISE 6 - PASSWORD POLICY
```

```
--1
```

```
CONNECT C##DBADMIN/Password1;

/*ERROR at line 1:
ORA-65140: invalid common profile name
when trying to creating the profile first attempt.. so to resolve this is says
common profile names must start with C## */

CREATE PROFILE C##studentProf LIMIT
  PASSWORD_LIFE_TIME 60
  PASSWORD_GRACE_TIME 10
  PASSWORD_REUSE_TIME 2
  PASSWORD_REUSE_MAX 2
  FAILED_LOGIN_ATTEMPTS 2
```

```

PASSWORD_LOCK_TIME 2
CPU_PER_CALL 3000
PRIVATE_SGA 500K
LOGICAL_READS_PER_CALL 1000;

--Profile created successfully

DESCRIBE DBA_PROFILES

SELECT * FROM DBA_PROFILES;
SELECT * FROM DBA_PROFILES WHERE PROFILE = 'C##STUDENTPROF';
--These are stored under the DBA_PROFILES which shows all profiles and limit

--2
--assign student B to profile created 'studentProf'

ALTER USER C##STUDENTB_1701497
PROFILE C##studentProf;

    --user altered

--3

DESC DBA_USERS;
--Show me the column names to specify a criteria for faster access and more useful
display

SELECT * FROM DBA_USERS
WHERE USERNAME = 'C##STUDENTB_1701497';

--4

    --WRONG PASSWORDS IN THE USER PASSWORD PARAMETER BELOW
CONNECT C##STUDENTB_1701497/&bdcj;
CONNECT C##STUDENTB_1701497/GGKFKDF;
CONNECT C##STUDENTB_1701497/DDDFDSD;
CONNECT C##STUDENTB_1701497/DDDFDSD;

/*ERROR:
ORA-28000: The account is locked.
*/
--The attempts has locked the account

--5

CONNECT C##DBADMIN/Password1;

ALTER USER C##STUDENTB_1701497 ACCOUNT UNLOCK;
--User account is now unlocked normal service resumed

--6

ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
IDENTIFIED BY WarSawPack;
/*ERROR at line 1:
ORA-28007: the password cannot be reused*/

```

```
ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
IDENTIFIED BY WarSawPack;
/*ERROR at line 1:
ORA-28007: the password cannot be reused*/
```

```
ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
IDENTIFIED BY WarSawPack;
/*ERROR at line 1:
ORA-28007: the password cannot be reused*/
```

```
SQL> CONNECT C##STUDENTA_1701497/pass1234;
Connected.
SQL> SELECT TABLE_NAME FROM USER_TABLES;

TABLE_NAME
-----
SALES

SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> SELECT TABLE_NAME , OWNER FROM ALL_TABLES WHERE OWNER='C##STUDENTA_1701497' ORDER BY TABLE_NAME;

TABLE_NAME
-----
OWNER
-----
SALES
C##STUDENTA_1701497

SQL> Select * from c##StudentA_1701497.Sales;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
A001              Apple              20          10
B002              Mango              40          70
C003              Pear              900         20

SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> REVOKE SELECT ON C##STUDENTA_1701497.SALES FROM C##STUDENTB_1701497;

Revoke succeeded.

SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> Select * from c##StudentA_1701497.Sales;
Select * from c##StudentA_1701497.Sales
*
ERROR at line 1:
ORA-00942: table or view does not exist
```

```
SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> CREATE VIEW VIEW_REORDER_1701497 AS
  2       SELECT * FROM SALES WHERE QUANTITY < 5 ;
CREATE VIEW VIEW_REORDER_1701497 AS
      *
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> GRANT CREATE VIEW TO C##STUDENT_1701497;

Grant succeeded.

SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> CREATE VIEW VIEW_REORDER_1701497 AS
  2       SELECT * FROM SALES WHERE QUANTITY < 5 ;

View created.

SQL>
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter user-name: C##DBADMIN
Enter password:
Last Successful login time: Sun Mar 13 2022 11:13:43 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> SELECT TABLE_NAME , OWNER FROM ALL_TABLES WHERE OWNER='C##STUDENT_1701497'
  2     ORDER BY TABLE_NAME;

no rows selected

SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> GRANT SELECT ON VIEW_REORDER_1701497 TO C##STUDENTB_1701497;

Grant succeeded.

SQL>
```

```
INHERIT PRIVILEGES

SQL> SELECT table_name, privilege FROM sys.dba_tab_privs WHERE grantor='C##STUDENTB_1701497';
TABLE_NAME
-----
PRIVILEGE
-----
C##STUDENTB_1701497
INHERIT PRIVILEGES

SQL>
SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> SELECT * FROM C##STUDENTB_1701497.VIEW_REORDER_1701497;

no rows selected

SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL>
```



```
SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> SELECT * FROM DBA_TAB_PRIVS WHERE GRANTEE = 'C##STUDENTB_1701497';

GRANTEE
-----
OWNER
-----
TABLE_NAME
-----
GRANTOR
-----
PRIVILEGE                                GRA HIE COM TYPE
-----
INH
---
C##STUDENTB_1701497

GRANTEE
-----
OWNER
-----
TABLE_NAME
-----
GRANTOR
-----
PRIVILEGE                                GRA HIE COM TYPE
-----
INH
---
C##STUDENTA_1701497

GRANTEE
-----
OWNER
-----
TABLE_NAME
-----
GRANTOR
-----
PRIVILEGE                                GRA HIE COM TYPE
-----
INH
---
VIEW_REORDER_1701497

GRANTEE
-----
OWNER
-----
TABLE_NAME
-----
GRANTOR
-----
PRIVILEGE                                GRA HIE COM TYPE
-----
INH
```

```
---  
  
SQL> CONNECT C##STUDENTB_1701497/pass1234;  
Connected.  
SQL> SELECT * FROM C##STUDENTA_1701497.VIEW_REORDER_1701497;  
  
no rows selected  
  
SQL> SELECT * from C##STUDENTA_1701497.Sales;  
SELECT * from C##STUDENTA_1701497.Sales  
                                *  
ERROR at line 1:  
ORA-00942: table or view does not exist  
  
SQL>
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Mar 14 05:32:36 2022  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Enter user-name: C##DBADMIN  
Enter password:  
Last Successful login time: Sun Mar 13 2022 12:59:47 -05:00  
  
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0  
  
SQL> ALTER SYSTEM SET audit_trail=db SCOPE=SPFILE;  
  
System altered.  
  
SQL>
```

```
SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> Shutdown immediate;
ORA-01031: insufficient privileges
SQL> Connect sys as sysdba
Enter password:
Connected.
SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> Startup
ORACLE instance started.

Total System Global Area 5133824504 bytes
Fixed Size                  9277944 bytes
Variable Size              956301312 bytes
Database Buffers          4160749568 bytes
Redo Buffers                7495680 bytes
Database mounted.
Database opened.
SQL> ;
```

```
SQL> CONNECT SYS/Backtrack#r3 AS SYSDBA;
Connected.
SQL> SHOW PARAMETER AUDIT;
```

NAME	TYPE	VALUE
audit_file_dest	string	C:\ORACLE19C\ADMIN\ORCL\ADUMP
audit_sys_operations	boolean	TRUE
audit_trail	string	DB •
unified_audit_sga_queue_size	integer	1048576
unified_audit_systemlog	boolean	FALSE

```
SQL>
```

```

SQL> CONNECT SYS/Backtrack#r3 AS SYSDBA;
Connected.
SQL> AUDIT SELECT TABLE, UPDATE TABLE, INSERT TABLE, DELETE TABLE BY C##STUDENT_1701497 BY ACCESS;
Audit succeeded.

SQL> SELECT * FROM DBA_STMT_AUDIT_OPTS;

USER_NAME
-----
PROXY_NAME
-----
AUDIT_OPTION          SUCCESS    FAILURE
-----
C##STUDENT_1701497

SELECT TABLE          BY ACCESS  BY ACCESS
C##STUDENT_1701497

INSERT TABLE          BY ACCESS  BY ACCESS
USER_NAME
-----
PROXY_NAME
-----
AUDIT_OPTION          SUCCESS    FAILURE
-----
C##STUDENT_1701497

UPDATE TABLE          BY ACCESS  BY ACCESS
C##STUDENT_1701497

USER_NAME
-----

```

```
SQL> GRANT CREATE ANY TABLE TO C##STUDENT_1701497;
```

```
Grant succeeded.
```

```
SQL> CONNECT C##STUDENT_1701497/pass1234;
```

```
Connected.
```

```
SQL> CREATE TABLE FOOD (  
 2     ID NUMBER,  
 3     Name VARCHAR(15)  
 4 );
```

```
Table created.
```

```
SQL> INSERT ALL  
 2     INTO FOOD VALUES (1001,'Apple')  
 3     INTO FOOD VALUES (1002,'Mango')  
 4     SELECT * FROM DUAL;
```

```
2 rows created.
```

```
SQL> SELECT * FROM FOOD;
```

```
      ID NAME  
-----  
    1001 Apple  
    1002 Mango
```

```
SQL> UPDATE FOOD  
 2     SET NAME = 'Chicken'  
 3     WHERE ID = 1002;
```

```
1 row updated.
```

```
SQL> DELETE FROM FOOD  
 2     WHERE ID = 1001;
```

```
1 row deleted.
```

```
SQL>
```

Query Result x						
All Rows Fetched: 5 in 0.16 seconds						
	USERNAME	TERMINAL	TIMESTAMP	OWNER	OBJ_NAME	ACTION_NAME
1	C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	INSERT
2	C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	INSERT
3	C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	SELECT
4	C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	UPDATE
5	C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	DELETE

```
SQL> CONNECT SYS/Backtrack#r3 AS SYSDBA
Connected.
SQL> NOAUDIT SELECT TABLE, UPDATE TABLE, INSERT TABLE, DELETE TABLE BY C##STUDENT_1701497;

Noaudit succeeded.

SQL>
```

```
SQL> CONNECT SYS/Backtrack#r3 AS SYSDBA
Connected.
SQL> NOAUDIT SELECT TABLE, UPDATE TABLE, INSERT TABLE, DELETE TABLE BY C##STUDENT_1701497;

Noaudit succeeded.

SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> INSERT ALL
  2 INTO FOOD VALUES (1003,'Cake')
  3 INTO FOOD VALUES (1004,'Bulla')
  4 SELECT * FROM DUAL;

2 rows created.

SQL> SELECT * FROM FOOD;

      ID NAME
-----
    1002 Chicken
    1003 Cake
    1004 Bulla

SQL> UPDATE FOOD
  2 SET NAME = 'Cheery Cake'
  3 WHERE ID = 1003;

1 row updated.

SQL> DELETE FROM FOOD
  2 WHERE ID = 1004;

1 row deleted.

SQL>
```

```
SELECT USERNAME, TERMINAL, TIMESTAMP, OWNER, OBJ_NAME, ACTION_NAME from DBA_AUDIT_TRAIL
WHERE USERNAME='C##STUDENT_1701497' AND OBJ_NAME = 'FOOD' ORDER BY TIMESTAMP;
```

Query Result x

SQL | All Rows Fetched: 5 in 0.012 seconds

USERNAME	TERMINAL	TIMESTAMP	OWNER	OBJ_NAME	ACTION_NAME
1 C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	INSERT
2 C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	INSERT
3 C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	SELECT
4 C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	UPDATE
5 C##STUDENT_1701497	PHILMORE	14/3/22	C##STUDENT_1701497	FOOD	DELETE

```
SQL> INSERT ALL
  2 INTO SALES VALUES ('D004','Cake',50,150)
  3 INTO SALES VALUES ('E004','Melon',1000,3)
  4 SELECT * FROM DUAL;
```

2 rows created.

```
SQL> Select * from Sales;
```

SALES_ID	PRODUCT	UNITCOST	QUANTITY
D004	Cake	50	150
E004	Melon	1000	3
A001	Apple	20	10
B002	Mango	40	70
C003	Pear	900	20

```
SQL> CONNECT C##DBADMIN/Password1;
Connected.
```

```
SQL> BEGIN
  2 DBMS_FGA.add_policy(
  3 object_schema => 'C##STUDENT_1701497',
  4 object_name => 'SALES',
  5 policy_name => 'QUANTITY_CHK_AUDIT',
  6 audit_condition => 'QUANTITY > 100',
  7 audit_column=> 'QUANTITY');
  8 END;
  9 /
```

PL/SQL procedure successfully completed.

```
SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> SELECT * FROM SALES WHERE QUANTITY < 8 ;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
E004              Melon              1000         3

SQL> CONNECT C##STUDENT_1701497/pass1234;
Connected.
SQL> SELECT * FROM SALES WHERE QUANTITY > 10 ;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
D004              Cake                50         150
B002              Mango               40          70
C003              Pear               900         20

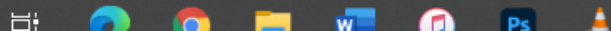
SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> SELECT DB_USER, SCN, SQL_TEXT, TIMESTAMP, POLICY_NAME FROM DBA_FGA_AUDIT_TRAIL;

DB_USER
-----
SCN
-----
SQL_TEXT
-----
TIMESTAMP
-----
POLICY_NAME
-----
C##STUDENT_1701497
5560583
SELECT * FROM SALES WHERE QUANTITY > 10

DB_USER
-----
SCN
-----
SQL_TEXT
-----
TIMESTAMP
-----
POLICY_NAME
-----
14-MAR-22
QUANTITY_CHK_AUDIT

SQL>
```

Type here to search




```

SQL>
SQL> CREATE OR REPLACE PROCEDURE InsertData_Sales(
  2   ID IN VARCHAR2 (25)
  3   , PRODUCT IN VARCHAR2 (25)
  4   , UNITCOST IN FLOAT
  5   , QUANTITY IN NUMBER
  6   ) AS
  7 BEGIN
  8   INSERT INTO SALES VALUES (ID, PRODUCT, UNITCOST, QUANTITY);
  9 END InsertData_Sales;
10 /
CREATE OR REPLACE PROCEDURE InsertData_Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL>

```

```

SQL Plus
SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> EXECUTE C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);
BEGIN C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5); END;
*
ERROR at line 1:
ORA-06550: line 1, column 7:
PLS-00201: identifier 'C##STUDENTA_1701497.SP_TIME_WINDOW_1701497' must be
declared
ORA-06550: line 1, column 7:
PL/SQL: Statement ignored

SQL> BEGIN
  2 C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);
  3 END;
  4 /
C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);
*
ERROR at line 2:
ORA-06550: line 2, column 1:
PLS-00201: identifier 'C##STUDENTA_1701497.SP_TIME_WINDOW_1701497' must be
declared
ORA-06550: line 2, column 1:
PL/SQL: Statement ignored

SQL> CONNECT C##DBADMIN/Password1;
Connected.
SQL> GRANT EXECUTE ON sp_time_window_1701497 TO C##STUDENTB_1701497;
GRANT EXECUTE ON sp_time_window_1701497 TO C##STUDENTB_1701497
*
ERROR at line 1:
ORA-04042: procedure, function, package, or package body does not exist

SQL> GRANT EXECUTE ON C##STUDENTA_1701497.sp_time_window_1701497 TO C##STUDENTB_1701497;
Grant succeeded.

SQL> CONNECT C##STUDENTB_1701497/pass1234;
Connected.
SQL> EXECUTE C##STUDENTA_1701497.sp_time_window_1701497('H010','CheesePatty',190,5);
PL/SQL procedure successfully completed.

SQL>

```

```

SQL Plus
SQL> Select * from tables
2 ;
Select * from tables
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> SELECT TABLE_NAME FROM USER_TABLES;

no rows selected

SQL> CONNECT C##STUDENTA_1701497/pass1234;
Connected.
SQL> SELECT TABLE_NAME FROM USER_TABLES;

TABLE_NAME
-----
FOOD
SALES

SQL> DESC SALES
Name                               Null?    Type
-----
SALES_ID                           VARCHAR2(25)
PRODUCT                            VARCHAR2(25)
UNITCOST                           FLOAT(126)
QUANTITY                           NUMBER(38)

SQL> CREATE TABLE Purchases(
2   p_id varchar2(25),
3   product varchar2(25),
4   quantity Number,
5   unitcost Float,
6   total_cost Float
7   );

Table created.

SQL> CREATE TABLE Purchases_log(
2   eventdate date default SYSDATE,
3   p_id varchar2(25),
4   description varchar2(50)
5   );

Table created.

SQL>

SQL> CREATE OR REPLACE TRIGGER Purchases_AUDIT
2   BEFORE INSERT ON Purchases
3   REFERENCING NEW AS NEW OLD AS OLD
4   FOR EACH ROW
5   BEGIN
6
7   INSERT INTO Purchases_log(p_id, description)
8   VALUES(:NEW.p_id,'INSERTING INTO PURCHASES');
9
10  UPDATE Purchases
11  SET total_cost = ((quantity * unitcost) + ((quantity * unitcost) * 0.165) )
12  WHERE p_id = :new.p_id;
13
14  END;
15  /

Trigger created.

SQL>

```

```

SQL Plus
23.3
P003          Cherry          4          5

SQL> SELECT * FROM PURCHASES;

P_ID          PRODUCT          QUANTITY  UNITCOST
-----
TOTAL_COST
-----
P001          Mango          2          20
P002          Cherry          4          5
23.3
P003          Cherry          4          5

SQL> SELECT * FROM Purchases_log;

EVENTDATE P_ID
-----
DESCRIPTION
-----
16-MAR-22 P001
INSERTING INTO PURCHASES

16-MAR-22 P002
INSERTING INTO PURCHASES

16-MAR-22 P003
INSERTING INTO PURCHASES

EVENTDATE P_ID
-----
DESCRIPTION
-----
16-MAR-22 P002
UPDATED TOTAL COST INTO PURCHASES

SQL> CREATE OR REPLACE TRIGGER Purchases_AUDIT
2  BEFORE DELETE ON Purchases
3  REFERENCING NEW AS NEW OLD AS OLD
4  FOR EACH ROW
5  BEGIN
6
7  INSERT INTO Purchases_log(p_id, description)
8  VALUES(:NEW.p_id,'DELETED FROM PURCHASES');
9
10 END;
11 /

Trigger created.

```

```

SQL Plus
LIMIT
-----
COM INH IMP
-----
C##STUDENTPROF
PASSWORD_VERIFY_FUNCTION      PASSWORD
DEFAULT
YES NO NO

PROFILE
-----
RESOURCE_NAME      RESOURCE
-----
LIMIT
-----
COM INH IMP
-----
C##STUDENTPROF
PASSWORD_LOCK_TIME      PASSWORD
2
YES NO NO

PROFILE
-----
RESOURCE_NAME      RESOURCE
-----
LIMIT
-----
COM INH IMP
-----
C##STUDENTPROF
PASSWORD_GRACE_TIME      PASSWORD
10
YES NO NO

PROFILE
-----
RESOURCE_NAME      RESOURCE
-----
LIMIT
-----
COM INH IMP
-----
C##STUDENTPROF
INACTIVE_ACCOUNT_TIME      PASSWORD
DEFAULT
YES NO NO

51 rows selected.

SQL> ALTER USER C##STUDENTB_1701497
  2  PROFILE C##studentProf;

User altered.

SQL>

```

```

SQL Plus
COM INH IMP
-----
C##STUDENTPROF
PASSWORD_GRACE_TIME          PASSWORD
10
YES NO NO

PROFILE
-----
RESOURCE_NAME                RESOURCE
-----
LIMIT
-----
COM INH IMP
-----
C##STUDENTPROF
INACTIVE_ACCOUNT_TIME        PASSWORD
DEFAULT
YES NO NO

51 rows selected.

SQL> ALTER USER C##STUDENTB_1701497
  2 PROFILE C##studentProf;

User altered.

SQL> SELECT USERNAME, PROFILE, ACCOUNT_STATUS FROM DBA_USERS WHERE USERNAME ='C##STUDENTB_1701497';

USERNAME
-----
PROFILE
-----
ACCOUNT_STATUS
-----
C##STUDENTB_1701497
C##STUDENTPROF
OPEN

SQL>
SQL> CONNECT C##STUDENTB_1701497/8%$bdcjj;
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> CONNECT C##STUDENTB_1701497/8%$bdcjj;
ERROR:
ORA-01017: invalid username/password; logon denied

SQL> CONNECT C##STUDENTB_1701497/8%$bdcjj;
ERROR:
ORA-28000: The account is locked.

SQL>

```

```

SQL Plus
User altered.

SQL> ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
  2 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
  2 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
  2 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 PASSWORD EXPIRE;

User altered.

SQL> ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL> ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack;
ALTER USER C##STUDENTB_1701497 IDENTIFIED BY WarSawPack
*
ERROR at line 1:
ORA-28007: the password cannot be reused

SQL>

```